

Stakeholder Engagement

Optimizing RAG Document Retrieval for Agronomic Advice, Team Baringo

1. Stakeholder groups selected

Rural agricultural extension workers and **women smallholder farmers**, both named directly in our Challenge 1 problem statement as key affected groups, but with different relationships to the system: extension workers are professional intermediaries who would use the tool to assist others; women farmers are often direct end-users with the least access to alternative information sources.

2. What matters most to them

Rural extension workers care most about **accuracy and trust**, a tool that surfaces the wrong or intent-mismatched document (e.g. prevention advice for a management question) could damage their professional credibility with the farmers they serve, so they want retrieval they can verify and correct, not a black box. They also value tools that reduce their workload rather than add to it. Women smallholder farmers prioritize **safe, respectful access to information**, given lower average access to formal education and frequent exclusion from formal agricultural decision-making spaces, they want a system that doesn't assume literacy, doesn't require navigating male-dominated channels, and treats their questions as legitimate.

3. Potential harms and benefits

Harms: For extension workers, being shown confidently-ranked but wrong documents (especially on underrepresented crops/regions, per our data audit) could lead them to unknowingly pass bad advice to farmers, eroding their authority. For women farmers, a text-only, English-only, formally-worded system risks excluding those with lower literacy or non-English fluency, effectively locking out the group our own problem statement identifies as most information-disadvantaged; there's also a risk the system reinforces existing gaps by only reflecting who already had a voice in the source documents.

Benefits: For extension workers, faster, more reliable retrieval expands how many farmers they can meaningfully support in a day, and having verifiable sources (per our transparency-by-attribution design) protects their credibility rather than undermining it. For women farmers, direct access to trustworthy agronomic information without needing to go through a male relative, cooperative leader, or extension officer is a meaningful form of autonomy and economic inclusion, particularly for decisions on their own plots.

4. Engagement methods selected (via the Action Catalogue)

- **Rural extension workers → Participatory Design.** They are domain experts already doing this work daily; the highest-value engagement is direct, iterative co-design where they review retrieval outputs and contribute real queries.
- **Women smallholder farmers → Distributed Dialogue.** This group is geographically dispersed, often time- and mobility-constrained, and may be underheard in centralized or mixed-gender settings, a method that runs small, local, parallel conversations (e.g. through existing women's farming groups) and synthesizes them centrally fits their actual constraints better than one central workshop.

5. Why each method fits

Participatory Design, Rural extension workers

Participatory design is an iterative, hands-on co-design approach in which the people who will actually use a system directly shape it, not just react to a finished product, but contribute examples, critique early outputs, and help define what "correct" even looks like. This fits rural extension workers precisely because they are not passive beneficiaries of this system; they are domain experts who already do the work our retrieval engine is meant to support. An extension worker fielding a farmer's question about bacterial leaf blight already has an intuitive sense of what a *correct* answer looks like, one they built through fieldwork, training, and direct exposure to the consequences of bad advice. That expertise is exactly the signal we most need and currently lack.

Concretely, participatory design sessions would work like this: extension workers bring real questions they've fielded from farmers (not our synthetic training queries, which we've already flagged as a limitation, they simulate realistic phrasing but aren't drawn from actual field interactions), and review our system's top-5 retrieved documents for each. Where the system surfaces a lexically-similar but wrong-intent document, the exact "manage vs. prevent" failure mode we found and partially fixed with cross-encoder reranking, an extension worker can immediately flag it in a way a graded relevance label alone can't fully capture, because they understand *why* it's wrong in the context of how that advice will actually be used in the field.

This method also directly addresses a risk we identified: an extension worker whose credibility depends on giving good advice will not trust, let alone adopt, a system that occasionally hands them a wrong answer with high apparent confidence. By involving them in design rather than only in eventual evaluation, we treat their professional judgment as an input to the system rather than a downstream check on it, which is also more respectful of their expertise and less likely to feel like the tool is trying to replace rather than support them, a concern we already flagged as a potential harm.

Participatory design is also well suited to the *iterative* nature of what we still have left to improve. Our own evaluation showed real, measurable failure modes (agro-zone-specific documents ranking lower than they should, crop/country coverage gaps), and a one-off survey or single feedback session wouldn't let extension workers meaningfully catch or correct these across iterations. Ongoing co-design sessions, reviewing new retrieval outputs, especially after coverage or model changes, makes it far more likely that their feedback

compounds into real improvements over time, not just a single round of comments that gets logged and forgotten. Because extension workers are a relatively small, identifiable, professionally-organized group (often through national extension services, several of which are already named sources in our corpus), it's also logistically realistic to sustain this kind of engagement over multiple sessions, unlike engaging farmers directly at scale.

Finally, this method builds a feedback loop that protects against exactly the kind of silent, confident failure we flagged as our most significant project risk: extension workers reviewing real outputs against real field cases are far more likely to catch a subtly wrong answer than any single aggregate accuracy metric would.

Distributed Dialogue, Women smallholder farmers

Distributed dialogue is an engagement method that runs the same structured conversation across many small, separate groups, often in different locations, at different times, rather than convening everyone into a single central event, and then synthesizes the results centrally into shared themes and recommendations. This method fits women smallholder farmers far better than a centralized workshop would, for reasons rooted directly in the access barriers our own problem statement already names.

First, geography and mobility: smallholder farmers, and women farmers in particular, are often unable to travel long distances for a single scheduled event, whether due to childcare responsibilities, farm labor demands, or limited transport access. A distributed model lets engagement happen inside existing local structures, women's farming cooperatives, community groups, savings circles, that already meet regularly in places these farmers can actually reach, rather than asking them to come to us.

Second, voice and safety: in many of the contexts our corpus covers, women are more likely to be excluded from mixed-gender public forums or formal decision-making spaces, or may simply speak less freely in front of men, including male relatives, community leaders, or extension officers. Running dialogue sessions within existing women's groups, facilitated by people already trusted within that space, creates the conditions for more honest feedback about what actually matters to them, safety concerns about how information is displayed, comfort with the phrasing of questions, trust in unfamiliar technology, than a single mixed, centralized session would surface.

Third, this method scales more realistically to the scope of who we're trying to reach. Our project is not meant to serve one village but a broad, dispersed population across multiple countries and language contexts. A distributed dialogue approach, many small, local conversations using a shared discussion guide, aggregated centrally, mirrors the shape of the actual population we're serving far better than a single representative panel could, and reduces the risk of over-indexing on whichever small group happened to be reachable for one event.

Concretely, this would mean working through existing women's farming groups (the kind already mentioned in our Challenge 1 problem statement as a target community) to run short, guided conversations about: how they currently seek agricultural information today, what would make them trust or distrust an automated

system, what language and literacy assumptions the system currently makes that don't match their reality, and whether the crop/region coverage of our corpus actually reflects what they grow. Centrally synthesizing these distributed conversations would let us identify patterns, for example, whether low trust concentrates around specific crops, regions, or types of questions, that we could act on directly, such as prioritizing corpus expansion or interface simplification.

Because this method meets women farmers where they already are, both geographically and socially, it treats their participation as something we need to actively make space for, rather than something we passively hope shows up if we just hold an open event.

6. How this would influence dataset and model design

Extension-worker participatory design would directly improve our evaluation data: real field queries (replacing our synthetic training questions) and expert-flagged retrieval failures would sharpen our hard-negative examples and reveal coverage gaps our own audit missed. Distributed dialogue with women's groups would likely push us toward non-English query support, simpler output phrasing, and a re-weighting of corpus coverage toward crops and regions currently underrepresented (e.g. Pearl millet, smaller countries), and toward being explicit about which answers are lower-confidence, rather than presenting every retrieved result with the same apparent authority.